



Machine Learning Operations Spring 2025

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1st Exam 27/06/2025 Version A

Name of the student:

Number:

Duration of the exam: **60 minutes**

In each question there exists information of how much each one is worth on a scale of **0 to 20 pts**. A multiple-choice wrong answer corresponds to a certain penalty. In each question you will see (score/ -penalty). Write the capital letter correspondent to the right answer.

1- For which process is the MLFlow fundamental? (1 pts/ -0.15pts)

- A) Creating versioned models.
- B) Creating virtual environments.
- C) Train models.
- D) Apply data unit tests.

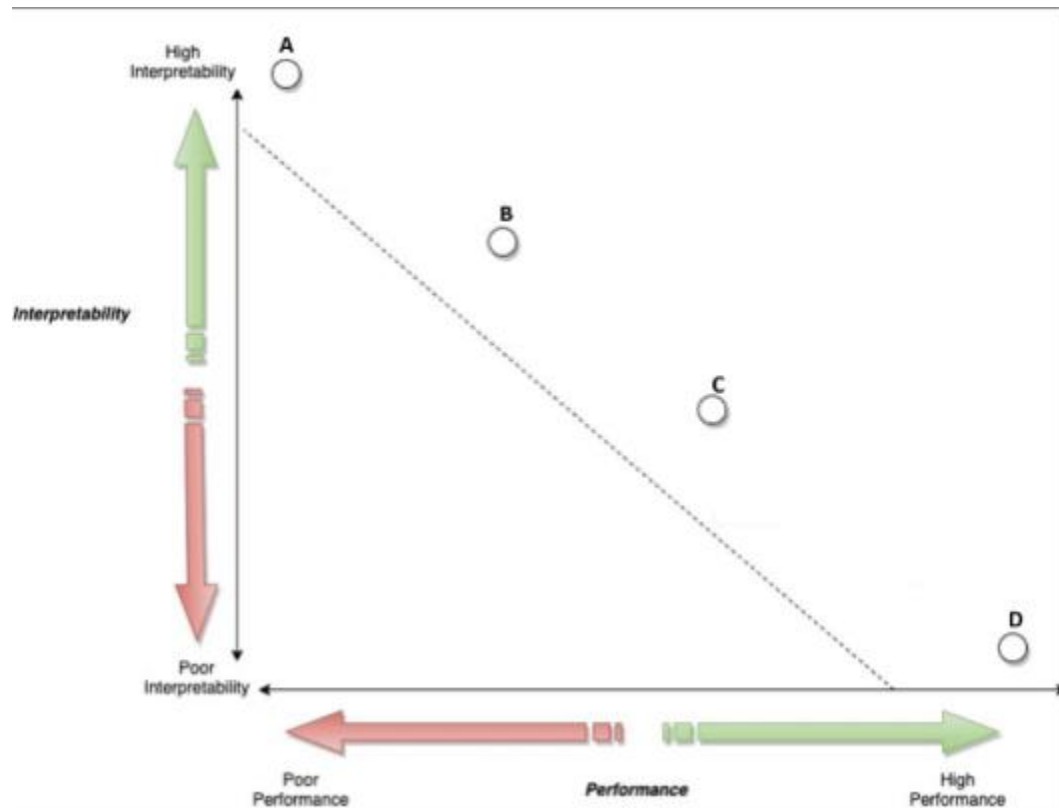
Choose the right option: ____ A

2- While having a model in production, we keep the development of new models, and we can put the new ones as challengers for the champion model. What are the advantages of shadow testing? (1 pts/ -0.15pts)

- A) We will test multiple models in different subpopulations, so that we need a statistical protocol.
- B) With shadow testing we save on the consumption of resources.
- C) We test against the same input, and it is safer since there is no room for sampling bias on the target events.

Choose the right option: ____ C

- 3- The Machine Learning models usually have a trade-off between interpretability and performance. Which option is the most correct to describe the following picture? (1 pts/ -0.15pts)



- A) A- Neural Networks, B- Linear Regression, C- Decision Trees, D- Ensemble Methods
- B) A- Ensemble Methods, B- Neural Networks, C- Decision Trees, D- Logistic Regression
- C) A- Linear Regression, B- Naive Bayes, C- Support Vector Machine, D- Neural Networks
- D) A- Naive Bayes, B- Linear Regression, C- Neural Networks, D- Ensemble Methods

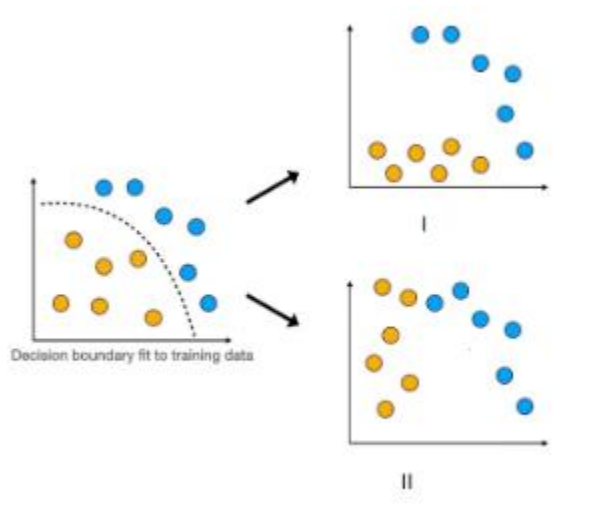
Choose the right option: ____ C

- 4- While having a model in production, we keep the development of new models and we can put the new ones as challengers for the champion model. When is it not possible to perform shadow testing? (1 pts/ -0.15pts)

- A) The ground truth cannot be evaluated for both models.
- B) The two models are different: one can be a tree model and the other is a neural network.
- C) The models have different features as inputs.

Choose the right option: ____A

- 5- In classification problems, after some time of the model being in production, we start to experience data drift. Sometimes there is an alert for retraining the model, or sometimes the drift is not enough to affect our performance metrics, and we can make small adjustments. The following figure represents two different situations of data drift for a classification problem. What option is correct? (1 pts/ -0.15pts)



- A) Both situations I and II confirm that we need to retrain the model.
- B) We do not need to retrain the model for both cases.
- C) In case I the performance metrics hold, so we do not need to retrain the model. In case II we need.
- D) In case II the performance metrics hold, so we do not need to retrain the model. In case I we need.

Choose the right option: ___ C

- 6- What kind of drift happens with Black Friday for ecommerce transactions? (1 pts/ -0.15pts)

- A) Sudden drift.
- B) Gradual drift.
- C) Incremental drift.
- D) Recurring concepts.

Choose the right option: ___ D

7- Kubernetes is used to orchestrate containers. (1 pts/ -0.15pts)

A) True

B) False

Choose the right option: ____ A

8- For which process is the **requirements.txt** fundamental? (1 pts/ -0.15pts)

A) Creating versioned models.

B) Creating virtual environments.

C) To define unitary data tests.

Choose the right option: ____ B

9- What solution from DevOps is one of the key features of MLOps general framework? (1 pts/ -0.15pts)

A) Implementation of software.

B) Continuous Integration and Continuous delivery.

C) Defined roles in the framework.

D) Model experimentations.

Choose the right option: ____ B

10- Which command in Git is used to upload your local repository's changes (commits) to a remote repository? (1 pts/ -0.15pts)

A) Push.

B) Pull.

C) Merge.

Choose the right option: ____ A

11- In what order are the following phases executed in a machine learning project?(1 pts/ -0.15pts)

I – Selection of ML algorithm

II – Data Exploration

III – Definition of the business use case

IV – Model monitoring

V – Model operationalization

VI – Model Development

A) I, II, III, IV, V, VI

B) III, II, I, VI, V, IV

C) II, III, I, VI, IV, V

D) II, I, III, VI, IV, V

Choose the right option: ____ B

12- What is the main difference between covariate shift and label drift? (1 pts/ -0.15pts)

A) Covariate shift is a consequence of label drift.

B) Covariate shift is related to a change in the distribution of the input variables of the model. The label drift corresponds to a change in the distribution of the target variable of the model.

C) Label drift is related to a change in the distribution of the input variables of the model. The covariate shift corresponds to a change in the distribution of the target variable of the model.

Choose the right option: ____ B

13- What are the costs of adopting a passive strategy to handle concept drift (1 pts/ -0.15pts)?

A) Costs of keeping the label at a high pace.

B) Costs of feedback systems and data drift solid tests.

C) Takes time to explore and find the optimal solution.

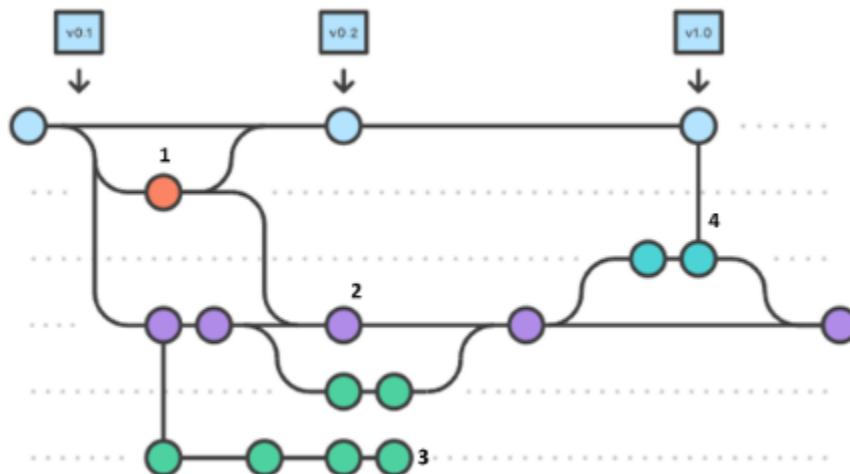
Choose the right option: ____ A

14- During our projects we used pytest to test the integrity of the functions. In which stage of our framework should it be included? (1 pts/ -0.15pts)

- A) Model experimentation.
- B) Continuous integration.
- C) Data unit tests.
- D) Monitoring.

Choose the right option: ____ B

15- When using Github, we apply several branch strategies to ensure the collaboration and development of our different pipelines between different team members to respect the Continuous Integration principles. Which option on the following image corresponds to the Hotfix branch? (1 pts/ -0.15pts)



- A) 2
- B) 3
- C) 1
- D) 4

Choose the right option: ____ C

16- What is the main difference between a container and a virtual machine(1 pts/ -0.15pts)

A) None. They are the same.

B) Containers virtualize the operating system instead of hardware, which makes them portable and efficient.

C) Virtual machines virtualize the operating system instead of hardware, which makes them portable and efficient.

D) Virtual machines use a requirements.txt, but containers do not use.

Choose the right option: ____ B

17- Permutation importance is a method used to explain my model at a local level ?(1 pts/ -0.15pts)

A) True.

B) False.

Choose the right option: ____ B

18- In the context of Kubernetes containers can be created and destroyed, not individual pods. (1 pts/ -0.15pts)

A) True.

B) False.

Choose the right option: ____ B

19- Model Governance is a new challenge in our present. What will have a great impact on the integration between Machine Learning governance and MLOps? (1 pts/ -0.15pts)

A) Size of the data.

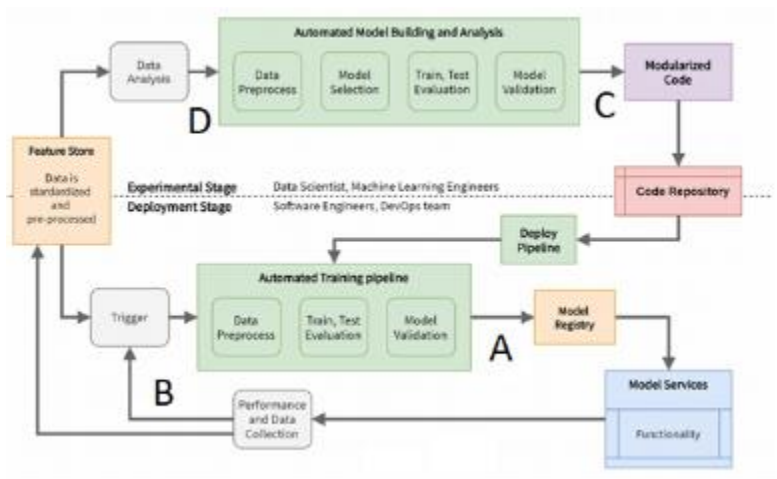
B) Regulations and number of ML models.

C) Type of software used to create models.

D) Feature selection method.

Choose the right option: ____ B

20- In the following figure, where should I include my data drift component on my pipeline? (1 pts/ -0.15pts)



A) A

B) B

C) C

D) D

Choose the right option: ____ B